

Diego A. Ortega

Software Engineer

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Summary

I build and operate production software end to end. For the past ten years I have been the only technical resource for a multi location dental practice, where I designed a custom remote desktop platform now running on 20 machines, automated recurring clinical and billing workflows, and own the entire technical environment from servers to endpoints. I work primarily in Python and JavaScript, with recent focus on local LLM deployment and offline AI tooling.

Skills

Programming: Python, Java, JavaScript, HTML/CSS

AI / Data: NumPy, Pandas, scikit-learn, model evaluation, local LLM deployment

Tools: Git/GitHub, VS Code, Linux, REST APIs

Domain: Healthcare IT, insurance claims and EOB workflows, HIPAA aware environments

Experience

Software Engineer, Sole Technical Lead

J. De La Osa DDS Inc. • Pomona, CA • 2016 to Present

- Architected and shipped GonzalezDesk, a custom remote desktop platform in Python deployed across 20 machines at multiple office locations, supporting LAN and cross location access over TCP/IP.
- Replaced all commercial remote access licensing by self hosting on organization owned infrastructure, retaining full data control and eliminating recurring per seat costs.
- Built a live remote update pipeline that pushes fixes and features to every deployed machine with no on site visits.
- Write and maintain Python automation for recurring clinical and administrative workflows.
- Work directly with dental insurance claims, EOB processing, and payer billing workflows.
- Sole engineer for the entire technical environment: servers, networking, endpoint management, backups, and vendor system integration.

Projects

GonzalezDesk

Python, TCP/IP Networking, Server Architecture, PyInstaller • Production

- Custom TCP/IP client and server handling screen transport, input relay, and cross location access via port forwarding.
- Remote update pipeline pushes new builds to all deployed machines with no on site visit required.
- Replaced commercial per seat licensing entirely while keeping all data on organization controlled hardware.

J.A.R.V.I.S v2

Python, Whisper, Piper TTS, Ollama, PyQt, Three.js • Personal build

- Real time voice pipeline using PyAudio and Whisper, with a custom Piper TTS audio chain featuring EQ, voice doubling, and reverb.
- Local LLM running through Ollama with persistent memory and user preferences stored in JSON.
- Holographic interface built in PyQt WebEngine and Three.js with animated bloom, particle systems, and layered rings.

AI Study Buddy

JavaScript, WebLLM, WebGPU, Local AI

- A fully local AI study assistant that runs real LLMs directly in the browser with WebGPU.
- Hardware check detects GPU and RAM and recommends the best model for the device.
- Subject and language modes with token by token local generation and cached model downloads.

NeuroForge

HTML, Python API, optional C++ engine

- Browser first AI workstation that turns prompts into structured concept frames, execution layers, launch plans, and risk reviews through a local Python backend.
- Multi panel interface with session history, backend status, metrics, and structured output cards.

Customer Churn Predictor 1.0

Pandas, scikit-learn, Flask • Archived

- Compared logistic regression, random forest, and gradient boosting; reported precision, recall, ROC AUC, and confusion matrix.
- Built a Flask API with an interactive web interface for predictions.

Earlier Frontend Work

SteemStore, UTube, Amazyn, Task Tracker, and Pixel Raiders. UI and game prototypes kept for reference.

Education

A.S. Computer Science

Mt. San Antonio College • In Progress

Planned transfer to California State University, Fullerton for B.S. Computer Science. Relevant coursework: Data Structures, Algorithms, Discrete Math, Databases, Operating Systems.

Languages

English and Spanish (fluent).